

YEAR 7 · SCIENCE

Q1 Review: Science Units 1 and 2

Science 1.1–1.4 — cells, specialised cells, tissues and organs

Science 2.1–2.4 — particles, changes of state, the water cycle

Science 2.5–2.8 — atoms, compounds, mixtures, acids and bases

Name: _____

Date given: _____

How to use this packet

Focus the basics · **Practice** like the assessment · **Challenge** the hardest — do not skip
Answer in full sentences. “Nucleus” is not a sentence.

Section A · Unit 1 — Cells, Tissues and Organs

A1 Every part has a job

FOCUS

Word help — two parts both “control” something

controls — is in charge of. **releases** — lets out. Read the words **after** “controls”: *what goes in and out* is one part; *everything the cell does* is another.

(a) Draw **one** line from each part of a cell to its job.

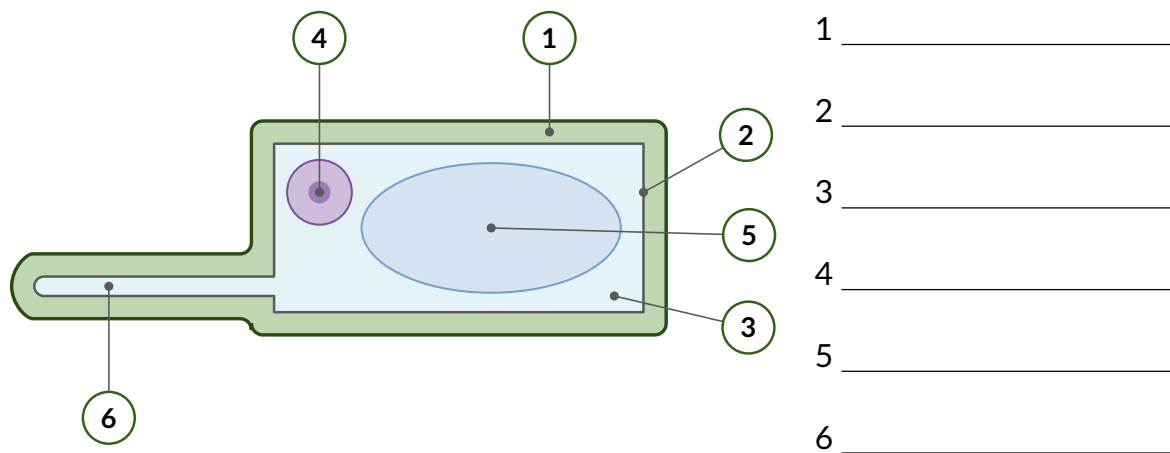
- | | |
|-----------------|---|
| cell membrane • | • makes food using sunlight |
| cytoplasm • | • releases energy from food |
| nucleus • | • holds a plant cell in shape |
| mitochondrion • | • controls what goes in and out of the cell |
| cell wall • | • is full of cell sap and keeps the cell firm |
| chloroplast • | • controls everything the cell does |
| sap vacuole • | • is where the cell's chemical reactions happen |

(b) Draw a ring around the **three** parts in (a) that an **animal** cell does **not** have.

A2 The root hair cell

PRACTICE

(a) Write the name of each numbered part.



(b) Water from the soil goes into the cell. Number these **1 to 4** in the order the water passes through them.

☐ cell membrane ☐ sap vacuole ☐ cell wall ☐ cytoplasm

(c) Why does this cell have a long, thin root hair?

The long root hair gives the cell ...

A3 Which level is it?

FOCUS

Each level is built out of the one before it

cell → **tissue** (similar cells, one job) → **organ** (several tissues) → **organ system** (organs, one shared job) → **organism**

Write each one in the correct column.

a neurone · the brain · ciliated epithelium · a mango tree · the digestive system
a root · the palisade layer · a red blood cell · the breathing system · a cat

Cell	Tissue	Organ	Organ system	Organism

A4 Mr Bowen's table has four mistakes**PRACTICE**

There is **one** mistake in each row. Cross it out and write the correct words next to it.

Cell	Function	Special structure
red blood cell	carries oxygen	has a very long axon
neurone	carries electrical signals	has a large sap vacuole
ciliated cell	absorbs water from the soil	has cilia (tiny hairs)
palisade cell	makes food by photosynthesis	has no nucleus

A5 Unit 1 puzzles**CHALLENGE****Word help**

side by side — in a line, touching. **squeezes** — pushes through a small space.

1. A typical human cell is 0.02 mm across.

(a) How many cells, side by side, make a line 1 mm long? _____

(b) How many make a line 1 cm long? _____

(c) A hair is 0.07 mm thick. About how many cells wide is it? _____

2. A white blood cell chases germs, squeezes between other cells and swallows the germs. Why would a cell wall be **no good** to it?

Section B · Unit 2 — Particles, Changes of State, the Water Cycle**B1 Draw the particles****FOCUS**

Draw about 12 particles in each box. Use circles of the same size.

**solid****liquid****gas**

B2 Solid, liquid or gas?

FOCUS

Tick every state it is true for. Some rows need two ticks.	Solid	Liquid	Gas
It keeps its own shape.			
It fills any closed container.			
It takes the shape of its container, but keeps the same volume.			
It can be poured.			
It cannot be compressed.			
The particles vibrate, but stay in the same place.			
The particles touch, but can move past one another.			

B3 Name the change

PRACTICE

Five changes of state

melting solid → liquid · **freezing** liquid → solid · **evaporating** and **boiling** liquid → gas
 · **condensing** gas → liquid

What happens	Change of state	Heating or cooling?
A bar of chocolate goes soft in a hot car.		
Mr Bowen breathes on a cold window. Drops of water appear.		
Melted butter goes hard in the fridge.		
A wet floor dries in the afternoon.		
Water in a kettle bubbles at 100 °C.		

B4 Explain it with particles

PRACTICE

(a) What happens to the particles when water boils?

Heat energy is transferred to ... The particles move ... Some of them escape ...

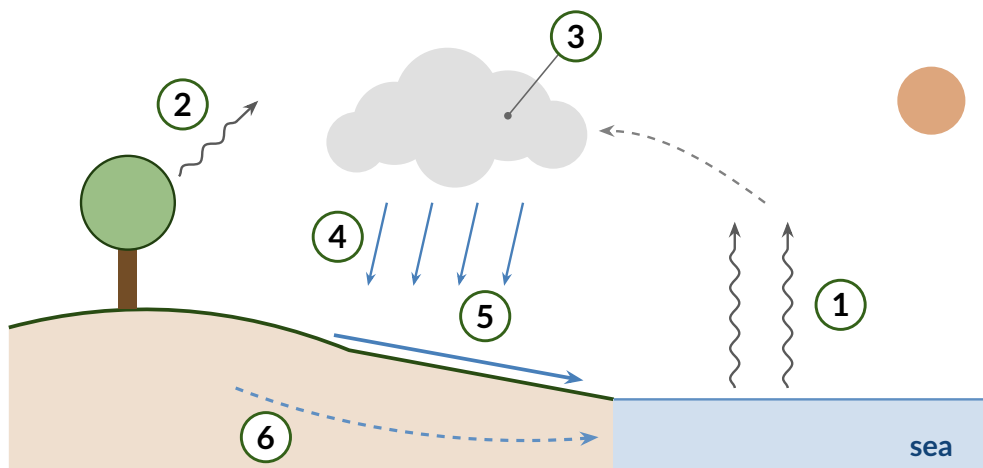
(b) Drops of water form on the outside of a cold can of drink. Where do they come from?
There is water vapour in ... When it touches the cold can, the particles ...

B5 The water cycle

PRACTICE

(a) Write the name of each numbered arrow. Use the words in the box.

condensation	evaporation	groundwater
precipitation	surface run-off	transpiration

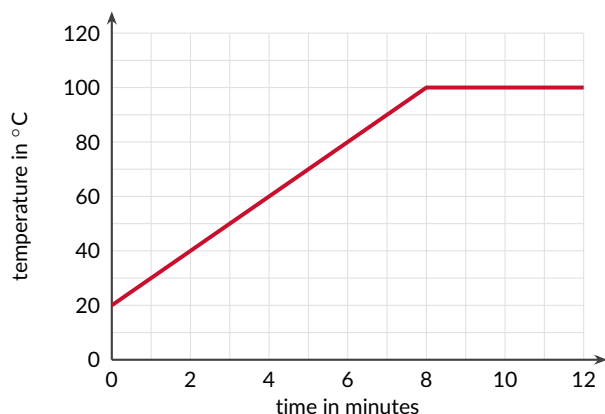


1 _____ 2 _____ 3 _____

4 _____ 5 _____ 6 _____

(b) You can never see water vapour. So what is a cloud made of?

(c) Rain is one kind of precipitation. Name **two** other kinds. _____

B6 Particle puzzles**CHALLENGE**

1. Mr Bowen heats a beaker of water for 12 minutes. The graph shows its temperature.

(a) When does the water start to boil?

(b) He is still heating it at 10 minutes. What is the temperature?

(c) Why does the temperature stop going up, even though he is still heating it?

2. Wet clothes dry faster on a hot, sunny day than on a cold day. Explain why, using particles.

Section C · Unit 2 – Atoms, Compounds, Mixtures, Acids and Bases

C1 Symbols, both ways**FOCUS**

The capital-letter rule

The first letter is **always a capital**. A second letter is **always small**: Mg, not MG or mg. A few come from Latin names: potassium is **K**.

(a) Write the symbol.

oxygen _____ potassium _____

magnesium _____ calcium _____

nitrogen _____ aluminium _____

(b) Write the name.

He _____ S _____

Li _____ Ar _____

F _____ Si _____

C2 Rows and columns

PRACTICE

	group							
	1	2	3	4	5	6	7	8
period 1				H				He
2	Li	Be	B	C	N	O	F	Ne
3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca						

 = metal

period = a **row** (across), not a lesson
group = a **column** (down), not people

(a) Period 3, group 2: _____

(b) Calcium is in period _____

(c) Two in group 8: _____

(d) Li and Na are in the same _____

C3 Symbols and formulae

PRACTICE

Reading a formula

The small number counts the atom **just before it**. No number means **one**. CO₂ is 1 carbon + 2 oxygen = 3 atoms.

K	Cl ₂	MgO	CH ₄	Ca	H ₂ O	NaOH	K ₂ CO ₃
---	-----------------	-----	-----------------	----	------------------	------	--------------------------------

Choose from the box.

(a) Write **two** elements made of single atoms. _____(b) Write the element whose particles are **two atoms** joined together. _____(c) Write the compound that is **cooking gas**. _____(d) Write the substance that contains a metal, carbon **and** oxygen, and give its name.
 _____ name: _____(e) How many atoms are in NaOH? _____ in K₂CO₃? _____

C4 Naming compounds

PRACTICE

Two naming rules

The **metal comes first**, and the non-metal ends in **-ide**: sodium + chlorine → sodium **chloride**. Two elements **plus oxygen** often end in **-ate**: calcium carbonate.

(a) Name the compound made from:

1. sodium and sulfur _____ 3. lithium and chlorine _____

2. oxygen and calcium _____ 4. magnesium, carbon and oxygen _____

(b) Which elements are in each compound?

1. potassium nitrate

2. copper sulfate

C5 Element, compound or mixture?

PRACTICE

(a) Each circle is an atom. Circles that touch are bonded. Write **E** (element), **C** (compound) or **M** (mixture) under each box.

1	2	3	4	5	6

(b) Draw **one** line from each word to what it means.

- | | |
|-------------------|---|
| element • | • contains only one substance |
| compound • | • different substances mixed together, but not bonded |
| mixture • | • made of only one kind of atom |
| pure • | • different kinds of atom bonded together |

(c) A mixture is easy to separate. How would you get:

the iron filings out of iron filings and sand?

the minerals out of mineral water?

C6 Acid or alkali?

FOCUS

Litmus

Blue litmus turns **red** in an acid. **Red** litmus turns **blue** in an alkali. Below 7 is acid; above 7 is alkali; 7 is neutral.

Write each one in the correct column.

tastes sour · pH 2 · pH 12 · turns blue litmus red · turns red litmus blue · tamarind
toothpaste · stomach acid · limewater · hydrochloric acid · sodium hydroxide · baking soda

Acid	Alkali

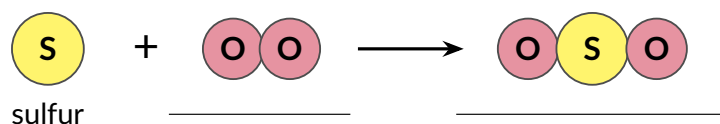
C7 Two questions from the end of the year

CHALLENGE

Word help

reactants — the substances you **start** with. **product** — the **new** substance that is made.
mean — add the results, then divide by how many there are.

1. Sulfur burns in oxygen. The picture shows the particles.



(a) Write the name of the other reactant, and of the product, under the picture.

(b) Write the formula of the product. _____

2. Mr Bowen has two bottles of alkali, **X** and **Y**. He puts 20 cm³ from one bottle in a flask with a few drops of universal indicator, and adds acid until it is neutralised. He does this three times for each bottle.

Bottle	Acid needed in cm ³			Mean
	1st try	2nd try	3rd try	
X	15	16	17	16
Y	23	26	23	

(a) How does he know when the alkali has been neutralised?

(b) Why does he do it three times, not once?

(c) Work out the mean for bottle **Y** and write it in the table.

(d) Which bottle holds the **stronger** alkali? Explain how you know.

Section D · Both Units

D1 Mr Bowen's lab notebook

CHALLENGE

Mr Bowen wrote this. It is full of mistakes. Find **six**: underline each one, then write the correct sentence.

Monday. I looked at my own cheek cells under the microscope. Each one had a cell wall and a big sap vacuole. Their mitochondria make food using sunlight.

Then I boiled some water for my tea. It boiled at 0 °C and turned into water vapour, which I could see as a white cloud above the kettle.

I put salt in my tea. Salt is a mixture of sodium and chlorine. I dipped red litmus paper in the salt water and it turned blue, so salt water is an acid. The tea was terrible.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Answer Key — Teacher Copy

A few question types here are also on the Quarter 1 Assessment, at your request; no item is. The list of which is kept with the assessment, not in this public file. A1, C3, C5(b), C6 and C7 are rewritten from the Cambridge end-of-year paper.

A1 Every part has a job

(a) membrane — in and out · cytoplasm — chemical reactions · nucleus — controls everything · mitochondrion — releases energy · wall — shape · chloroplast — food using sunlight · sap vacuole — cell sap, firm
(b) cell wall, chloroplast, sap vacuole

The membrane and the nucleus swapped is the English trap the word help names: both jobs start with “controls”. Mitochondrion to “makes food” means energy and food have become one idea — the chloroplast *makes* food, the mitochondrion *releases energy* from it.

A2 The root hair cell

(a) 1 cell wall 2 cell membrane 3 cytoplasm 4 nucleus 5 sap vacuole 6 root hair
(b) cell wall 1, cell membrane 2, cytoplasm 3, sap vacuole 4
(c) It gives the cell a big surface, so water can move in easily / it absorbs more water from the soil.

(a) 1 and 2 swapped: the wall is the thick green band outside, the membrane the dark line just inside it.
(b) is the 1.3 book question (p. 20 Q4) that no packet had asked. (c) “to absorb water” is the function, not how the shape helps — push for “big surface”.

A3 Which level is it?

Cell: neurone, red blood cell · Tissue: ciliated epithelium, palisade layer · Organ: brain, root · Organ system: digestive, breathing · Organism: mango tree, cat

The palisade layer (tissue, not cell) and the root (organ — 1.4 named roots and flowers as organs) are the two traps.

A4 Mr Bowen’s table

red blood cell: *very long axon* → full of haemoglobin / no nucleus · neurone: *large sap vacuole* → a very long axon · ciliated cell: *absorbs water from the soil* → sweeps mucus out of the airways · palisade cell: *has no nucleus* → packed with chloroplasts

Row 4 is the hard one: “no nucleus” is a real adaptation, just of a different cell. A student who crosses out the function in row 1 or 2 has not read the column headings.

A5 Unit 1 puzzles

1. (a) $1 \div 0.02 = 50$ (b) **500** (c) $0.07 \div 0.02 = 3.5$, so **about 3 or 4**
2. A cell wall is stiff and holds the cell in one shape, so the white blood cell could not change shape to squeeze between cells or wrap around germs.
1 is Unit 3 maths: $0.02 \times 50 = 1$. An answer of 0.5 or 5 means the digits moved the wrong way. 2 is the 1.3 exit question, turned round: a student who writes “it has no cell wall” has described the cell, not answered why.

B1 Draw the particles

Solid: touching, in regular rows. Liquid: touching, no pattern. Gas: far apart, no pattern. Circles the same size in all three.

B2 Solid, liquid or gas?

own shape: S · fills any container: G · shape of container, same volume: L · can be poured: L and G · cannot be compressed: S and L · vibrate in place: S · touch but move past: L

“Can be poured” and “cannot be compressed” are the two-tick rows; a single tick on either is the “only” idea not yet understood.

B3 Name the change

chocolate: melting, heating · breath on a window: condensing, cooling · butter in the fridge: freezing, cooling · wet floor: evaporating, heating (from the air around it) · kettle at 100 °C: boiling, heating

“Freezing” for butter sounds wrong to the class because the fridge is not below zero — say so: freezing is liquid to solid, at whatever temperature that substance does it. Accept “heating” or “neither” for the floor if explained.

B4 Explain it with particles

(a) Heat energy is transferred to the particles. They move faster and faster. Some escape the attractive forces and leave as a gas (steam / water vapour).

(b) There is water vapour in the air. When it touches the cold can, the particles lose energy and slow down, and the forces pull them together into a liquid: it condenses.

(b) “the water comes through the can” is the misconception: the can is sealed.

B5 The water cycle

1 evaporation 2 transpiration 3 condensation 4 precipitation 5 surface run-off 6 groundwater

(b) tiny drops of liquid water (c) any two of snow, hail, sleet

(b) “water vapour” is the 2.4 misconception exactly — a cloud is drops that have *already* condensed. 1 and 2 swapped: transpiration is water leaving a plant.

B6 Particle puzzles

1. (a) at 8 minutes (at 100 °C) (b) 100 °C (c) The heat is changing the liquid into a gas instead of raising the temperature.

2. The particles in the water get more energy when it is hot, so they move faster and more of them escape (evaporate) from the surface.

1(b) about 120 °C comes from carrying the line on — the trap the graph is for. 2: “the sun dries them” is not a particle answer.

C1 Symbols, both ways

(a) O, K, Mg, Ca, N, Al (b) helium, sulfur, lithium, argon, fluorine, silicon

MG, mg or CA: the capital-letter rule. P for potassium is the Latin trap — P is phosphorus.

C2 Rows and columns

(a) Mg (b) 4 (c) any two of He, Ne, Ar (d) group

(a) Al or Be: period and group swapped.

C3 Symbols and formulae

- (a) K and Ca (b) Cl_2 (c) CH_4 (d) K_2CO_3 , potassium carbonate (e) 3 and 6
 (b) is an element even though it has a 2 in it – the hw07 E4 mistake again. (e) 5 for K_2CO_3 means the 2 was not counted.

C4 Naming compounds

- (a) sodium sulfide, calcium oxide, lithium chloride, magnesium carbonate
 (b) potassium, nitrogen, oxygen · copper, sulfur, oxygen
 (a) 2 is written oxygen-first on purpose: “oxide calcium” means the student copied the order instead of putting the metal first.

C5 Element, compound or mixture?

- (a) 1 E 2 E 3 C 4 M 5 M 6 C
 (b) element – one kind of atom · compound – different atoms bonded · mixture – mixed but not bonded · pure – only one substance
 (c) with a magnet (iron is magnetic, sand is not) · evaporate the water away (heat it in an evaporating basin)
 Box 2 is the trap: atoms joined in pairs, but all one kind, so an element (like O_2). Box 6 is one substance even though it has two kinds of atom.

C6 Acid or alkali?

Acid: tastes sour, pH 2, turns blue litmus red, tamarind, stomach acid, hydrochloric acid
 Alkali: pH 12, turns red litmus blue, toothpaste, limewater, sodium hydroxide, baking soda

C7 Two questions from the end of the year

1. (a) reactants: sulfur, oxygen; product: sulfur dioxide (b) SO_2
 2. (a) the universal indicator shows pH 7 (the neutral colour) (b) to check the results / make them reliable / spot a mistake (c) $(23 + 26 + 23) \div 3 = 72 \div 3 = 24$ (d) Y: it needed more acid to neutralise the same 20 cm^3 .
 2(d) “X, because 16 is a smaller number” is the misreading to expect: the more acid it takes, the stronger the alkali. Do not ask what is made – the products are Stage 8.

D1 Mr Bowen’s lab notebook

- Cheek cells are animal cells: they have no cell wall ...
- ...and no (large) sap vacuole.
- Mitochondria release energy from food; chloroplasts make food (and cheek cells have none).
- Water boils at 100°C , not 0°C (it freezes / melts at 0).
- You cannot see water vapour; the white cloud is tiny drops of liquid water.
- Salt is a compound (sodium chloride), not a mixture. *Also accept:* red litmus turning blue means an alkali, not an acid (salt water is neutral).

There are seven errors in the text on purpose, so a student who finds a different six is not penalised. “The tea was terrible” is not a mistake.